

THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD

Department of Software Engineering

LAB 03

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Q1: Sequence diagram for cash withdrawal from an ATM

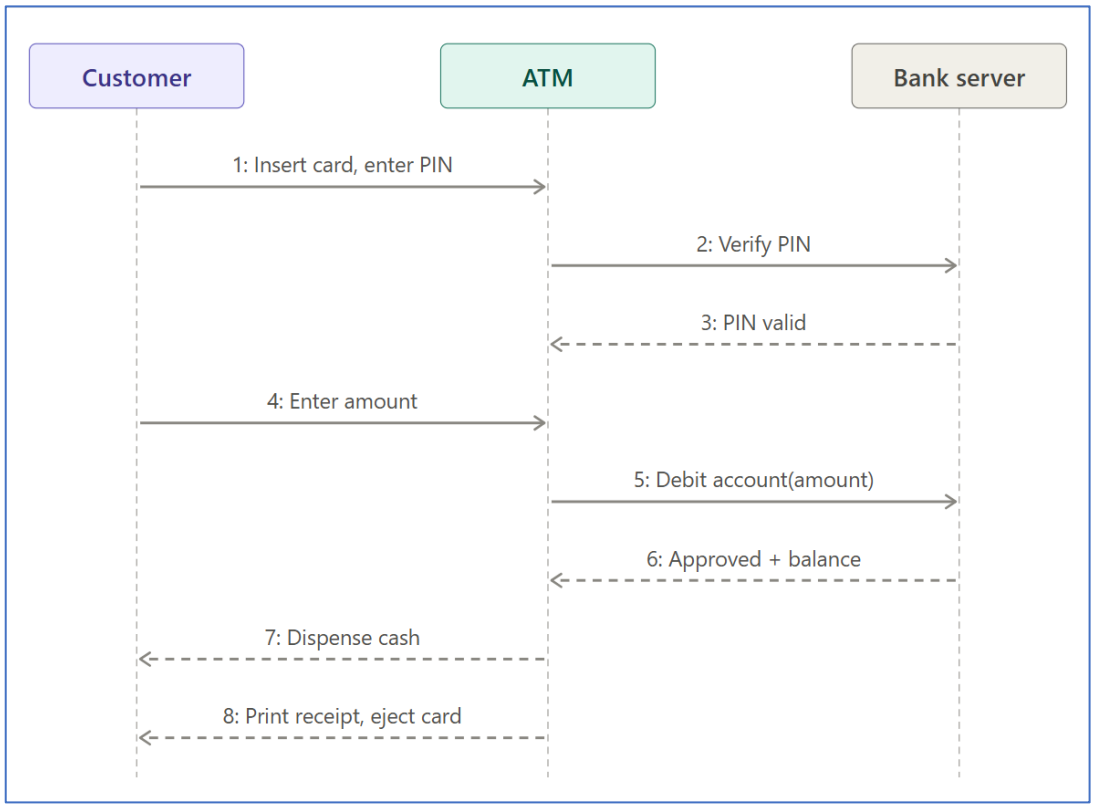


Figure 1Sequence diagram for cash withdrawal from an ATM

Participants (lifelines): ***Customer, ATM, Bank server.*** A solid arrow is a request, and a dashed arrow is a reply.

1. The Customer inserts the card and enters the PIN.
2. The ATM sends the PIN to the Bank server for verification.
3. The Bank server replies that the PIN is valid.
4. The Customer enters the amount to withdraw.
5. The ATM asks the Bank server to debit the account. The Bank checks the balance and deducts the amount.
6. The Bank server sends back an approval along with the new balance.
7. The ATM dispenses the cash to the Customer.
8. The ATM prints the receipt and ejects the card.

Alternate flow: If the PIN is wrong or the balance is insufficient, the Bank server sends a rejection. The ATM then shows an error message and returns the card. In UML, this is drawn with an alt fragment.

Q2: Sequence diagram for the online ticket reservation system

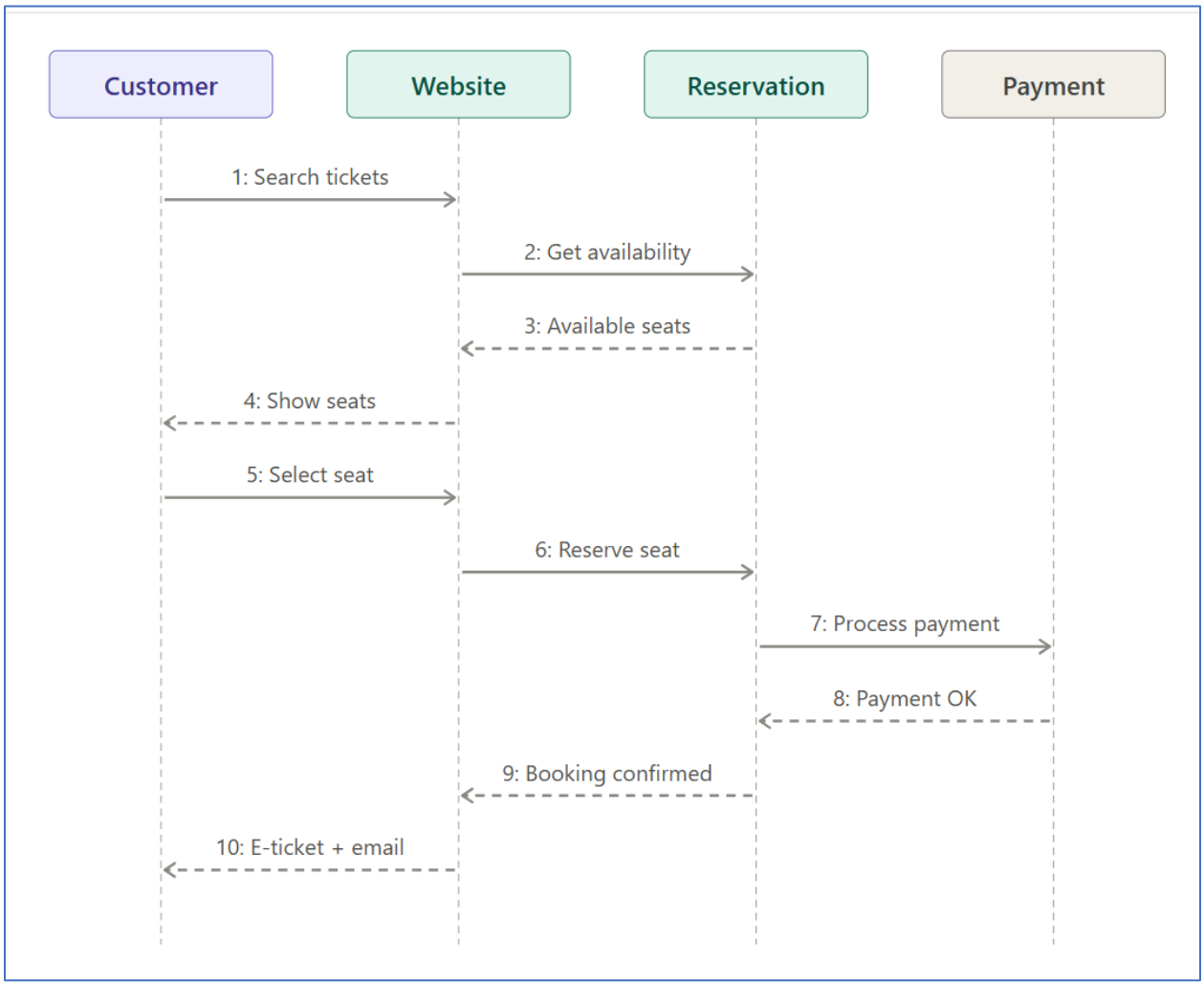


Figure 2Q2: Sequence diagram for the online ticket reservation system

Participants (lifelines): ***Customer, Website, Reservation system, Payment gateway.***

1. The Customer searches for tickets on the website (event, date, route, etc.).
2. The Website requests availability from the Reservation system.

3. The Reservation system returns the list of available seats.
4. The Website displays the available seats to the Customer.
5. The Customer selects a seat.
6. The Website asks the Reservation system to reserve the seat, and the seat is temporarily locked.
7. The Reservation system sends a payment request to the Payment gateway.
8. The Payment gateway confirms that the payment was successful.
9. The Reservation system confirms the booking to the Website.
10. The Website shows the e-ticket to the Customer and sends it by email.

Alternate flow: If the payment fails, the Reservation system releases the seat lock and the Customer sees a “payment failed” message.